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Remote sensing images super-resolution with deep convolution networks

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Abstract

Remote sensing image data have been widely applied in many applications, such as agriculture, military, and land use. It is difficult to obtain remote sensing images in both high spatial and spectral resolutions due to the limitation of implements in image acquisition and the law of energy conservation. Super-resolution (SR) is a technique to improve the resolution from a low-resolution (LR) to a high-resolution (HR). In this paper, a novel deep convolution network (DCN) SR method (SRDCN) is proposed. Based on hierarchical architectures, the proposed SRDCN learns an end-to-end mapping function to reconstruct an HR image from its LR version; furthermore, extensions of SRDCN based on residual learning and multi scale version are investigated for further improvement,namely Developed SRDCN(DSRDCN) and Extensive SRDCN(ESRDCN). Experimental results using different types of remote sensing data (e.g., multispectral and hyperspectral) demonstrate that the proposed methods outperform the traditional sparse representation based methods.

Keywords Remote sensing imagery · Super-resolution · Convolution neural network

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1 Introduction

Spatial resolution is one of the key factors for remote sensing images [38] since it can greatly impact the performance of practical applications, such as target detection, pattern classification [28], etc. Super-resolution (SR) methods improve the spatial resolution of images from low-resolution (LR)to high-resolution (HR).

From the imaging theory, SR can be regarded as an inverse solution of blurring and down sampling.Various methods for image SR have been proposed, such as upsampling, edge enhancement and interpolation-based methods [17]. Tsai and Huang carried out a frequency domain analysis to solve the SR problem [31]. Discrete cosine transform (DCT) based methods have been employed to handle the noise problem [25]. In [16], an MRF model based approach was introduced to generate super-resolution land cover maps from remote sensing data. Spatial domain approaches have also been developed, such as maximum a posteriori [26], edge-directed interpolation [22, 32]. Recently, dictionary-based or learning mapping function methods were introduced in [4, 8, 9, 34]. Li et al. proposed a prior model based on the MAP for reconstructing an HR image from a sequence of LR remote sensing images [21]. In order to solve the problem that the interpolated image may lack the high frequencies, Zhang et al. used a support vector regression (SVR) method which performs nonlinear mapping to learn the prior model between LR and HR image patches [37]. Dong et al. [7] directly learned an end-to-end mapping between the low and high-resolution images. Yuan et al. [35] proposed a new framework to enhance the resolution of hyperspectral images by exploiting the knowledge from natural images. In [24], shallow and deep convolution neural networks (CNNs) were tested for Landsat image super-resolution enhancement, trained using Sentinel-2, in three study sites representing boreal forest, tundra, and cropland or woodland environments.

Sparse representation has also been introduced to super-resolution. A nonlocal pairwise dictionary was employed in the field of remote sensing image SR [11]. A pair sparsity model based sparse dictionary has been proposed [39]. Sparse coding super resolution (SCSR) was presented in [10]. A method based on the compressive sensing, structural self-similarity and dictionary learning was proposed [23] aiming to identify dictionary with sparse coding. In [5], SR using multihypothesis prediction (MHSR) was proposed.Among these approaches, the relevance between HR and LR images was learned from dataset of HR-LR pairs through sparse coding algorithm, and then applied to recover the HR image. The learning-based methods have improved the performance obviously; however, it needs a lot of computing resource in both procedures of training the dictionaries and reconstructing the HR image.

Deep convolutional neural networks (CNN) is of great interest recently, which was firstly proposed by Geoffrey Hinton [14], and then applied in the field of image classification [12, 19, 27]. With the development of GPUs (Graphics processing unit) and NPUs (Neural Processing Units), the parallel computing capabilities have improved greatly, making CNNs deeper and faster. In [6], an end-to-end deep learning method mapping between the low and high resolution images. In [33], the question whether large-capacity and data-driven models have become the dominant solution to the ill-posed super-resolution problem was arised. They combine domain expertise represented by the conventional sparse coding model with the key ingredients of deep learning. Inspired by VGG used for ImageNet classification, a very deep convolutional network was used for single image SR [18]. Despite using deep learing have a breakthroughs in SR, when we super-resolve at large upscaling factors, one unresolved problem how do we recover the finer texture details are faced. Ledig et al. [20] utilize a generative adversarial network (GAN) for image SR to resovle the problem above. Zeng et al. [36] removed the bicubic interpolation operation which is handcraft up-sampling

and not intelligent enough and explored the potential of deep neural networks in single image SR.

Deep CNN has been applied in many fields, such as classification [1, 13, 19, 27] and denoising [2, 15]. In this work, motivated by these aforementioned studies and impressive achievements of deep learning, a super-resolution with deep convolution network (SRDCN) is proposed.SRDCN learns an end-to-end mapping between the HR and LR images, and combines the capacity of feature extraction and edge enhancement in all hierarchical layers. SRDCN does not involve any optimization problems and its structure is quite simple. Based on SRDCN, a residual learning based method (DSRDCN) is investigated. Inspired by semantic segmentation and Inception proposed by Google [29, 30], an extensive SRDCN (ESRDCN) is also designed for SR purpose. Compared to the state-of-art dictionary learning-based methods, deep CNN based methods only need some training data for tuning network parameters, and take much less time to recover an HR image.

The performance of our proposed methods (SRDCN, DSRDCN and ESRDCN) are compared with bicubic interpolation, and dictionary based methods SCSR and MHSR. Results show that the improved versions developed based on SRDCN with residual learning and multi-scale performs much better in recovery quality without increasing any computational complexity. Classification results with the SR images also show higher accuracies with proposed methods.

The paper is organized as follows. Section 2 describes the proposed methods in details. Section 3 demonstrates the experimental performance and analysis. Section 4 makes some concluding remarks.

2 Super-resolution techniques for remote sensing images

SR is an ill-posed problem, which aims to reconstruct an HR image $\mathbf{I}^{HR}$ from a given LR image $\mathbf{I}^{LR}$. Motivated by recent compressive sensing theories [3] and sparsity prior, sparse coding super-resolution (SCSR) was presented in [10]. The observed image $\mathbf{I}^{LR}$ can be obtained from the HR version $\mathbf{I}^{HR}$ by downsampling (L) and blurring (H), and the model function can be expressed as,

$$\mathbf{I}^{LR} = \mathbf{LHI}^{HR}, \tag{1}$$

2.1 Super-resolution with deep convolution network

The proposed SRDCN method learns an end-to-end mapping function between HR and LR image patches. Figure 1 illustrates the overall framework of the proposed SRDCN. Since the LR image is smaller than the objective output HR one, the input LR image is firstly upscaled to the desired size with linear interpolation [17].The input image $\mathbf{I}^{LR}$ is fed to the

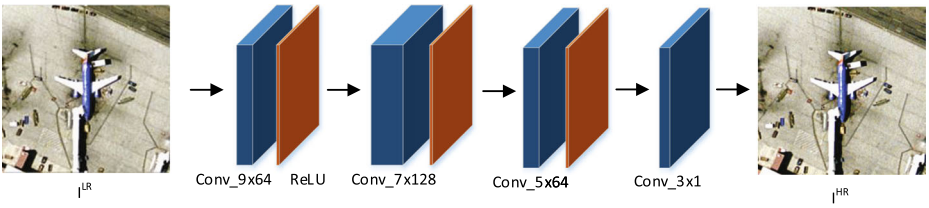

Fig. 1 The architecture of proposed SRDCN

network to recover the HR image $\mathbf{I}^{HR}$. We formulate the procedure as $F(\mathbf{X})$, which consists of four parts:

- **Low-level Feature Extraction:** This step is designed to extract all represents from input $\mathbf{I}^{LR}$ and expressed as a series of feature maps. Each pixel is converted to vector.
- **High-level Feature Extraction:** In this operation, the feature maps from previous step is transformed to another high dimension vector. The convolution operation can extract feature from previous feature maps.
- **Features Mapping and Reduction:** In order to reduce the hyper parameters, the operation is designed to mapping the high dimension vectors to a low one.
- **Reconstruction:** this operation can flatten all the information to reconstruct a high resolution image I^{HR}.

The main task of **Low-level Feature Extraction**, **High-level Feature Extraction** and **Features Mapping and Reduction** procedures are to collect compositions from the input $\mathbf{I}^{LR}$ that are fed into the CNN network and express all information into feature maps. It is noteworthy that the corner, edge and line can be dig out from the feature maps. These procedures can be formulated as:

$$\mathbf{Z}^l = ReLU(\mathbf{Z}^{l-1} * \mathbf{W}^l(+\mathbf{B}^l)) \tag{2}$$

Where the $\mathbf{Z}^l$ is the feature map extracted in the lth layer. The $\mathbf{W}$ and $\mathbf{B}$ are the weights and bias parametes in the convolution layer. The *ReLU* (rectified linear unit) can help sparsity of network and remove the negative features. And in **Features Mapping and Reduction** the bias $\mathbf{B}$ is not involved, which can be treated as mapping operation. While the *ReLU* is not applied in phase of **Reconstruction**, which can be formulated as.

$$\mathbf{I}^{HR} = \mathbf{Z} * \mathbf{W} + \mathbf{B} \tag{3}$$

The loss function is expressed with Euclidean (L_2) distance between the target $\mathbf{y}^i$ and prediction $F(\mathbf{x}^i)$.

$$L(\Theta, \mathbf{x}) = \frac{1}{N}\sum_{i=1}^{N}\left\| F(\mathbf{x}^i) - \mathbf{y}^i \right\|_2^2, \tag{4}$$

Where N denotes the number of training samples. The loss function is minimized using stochastic gradient descent with the backpropagation. The L_2 loss favors a high PSNR (Peak signal-to-noise ratio). The PSNR is a widely-used metric for quantitatively evaluating image denoise and reconstruction quality.

2.2 Super-resolution with deep convolution network based residual learning

With the assumption that HR and LR images are highly similar, and inspired by the residual learning techniques of RESNET [13] for image classification, an improved version based on the proposed SRDCN with residual learning is investigated, named as DSRDCN. DSRDCN attempts to utilize the detailed difference information to enhance the high frequency components.

Figure 2 illustrates the architecture of the proposed DSRDCN. The architecture contains more layers with a shortcut. The shortcut pass the feature map extracted from the first layer to subsequent operation,which can keep the context of the full image. With this shortcut, more high level dimension features can be extracted with increased layers. Besides, the shortcut can reuse the feature, which can help different level dimension fusion. Thus,

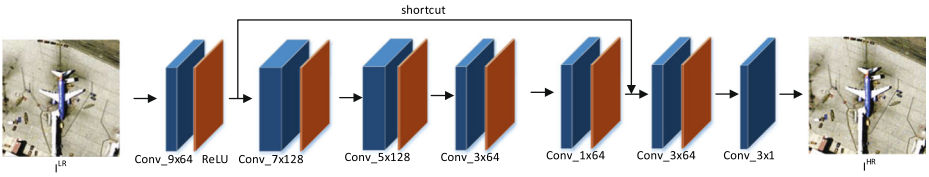

Fig. 2 The architecture of proposed DSRDCN

the network gets deeper with the residual learning. This procedure can be wrote as the formulation below

$$\mathbf{Z} = \mathbf{F}(\mathbf{X}) + \mathbf{X} \tag{5}$$

The shortcut in Fig. 2 pass the feature to the subsequent layer. While in the main path of the module, the feature map focus more on the local information. The F(X) in the (5) means the residual mapping.

2.3 Extensive super-resolution with deep convolution network

Inspired by the framework of Inception [29] and the relation between received field and the size of convolution kernel. An external CNN block is designed for our super resolution assignment. Three different sizes of convolution kernels are applied to extracted features from the same domain. As we know, different sizes of kernel mean different received field. So diverse feature maps are combined through concatenation. The detail architecture of network is illustrated in Fig. 3. Predicting a center part pixels correctly requires more surrounding pixels to improve the continuity, a larger kernel can help improve quality of reconstruction. A multiple scale of feature maps contain more information from a single region.

2.4 Network training

In our experiments, the LR images are first downsample and blurred from HR images, which are as the training target. The size of training patch is set as $f_{in} = 64$, and the target patches are same with the size of f_{in}. The dataset is separated into training S_{train} and validation S_{val}. The target image patches I^{HR} is prepared as $f_{in} \times f_{in} \times 1$. And the patches feed into the network are blurred with a Gaussian kernel and down-sample with a scale of 2 via nearest interpolation, and upsampled with same scales via bicubic interpolation. There are

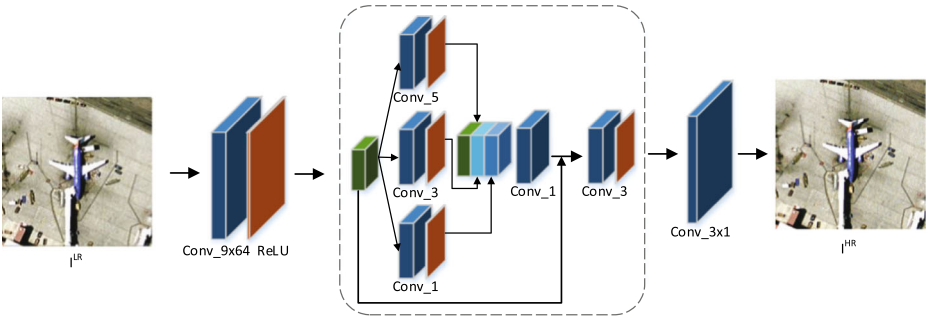

Fig. 3 The architecture of proposed ESRDCN

very few paired HR and LR images for the training dataset in all the experiments, so for data augmentation in the train phase, the flip and rotation operations are applied to the image from the training dataset.

During network training, the backpropagation algorithm is used to minimize the loss function and find the optimal hyper parameter of network. And the Stochastic gradient descent (SGD) is employed in our experiments. In order to achieve the objective of finding the optimal parameters, $\Theta = \{\mathbf{W}, \mathbf{B}\}$ is defined as the hyper parameters set to minimize the loss function. All the weights and biases are initialized with gaussian normalization at the first training phase. And, all the above three network are trained on the same dataset.

3 Experiments

In this section the proposed methods are applied to two hyperspectral image datasets to assess the super resolution and SR classification abilities. Besides, natural images (RGB) and multi-spectral remote sensing images (7 bands) are also tested for SR performances. To evaluate the results objectively, the PSNR (peak signal-to-noise ratio), RMSE (root mean square error), SSIM (structural similarity) and elapsed time are used as the quantitative metrics. Comparisons with other state-of-the-art methods are also conducted. All the experiments are implemented with *Python* on *PyTorch*[1], and conducted on the *Ubuntu 14.04*, dual *Intel E5 2683*, 32-GB memory with GPU of *Nvidia GTX 1080*.

3.1 Experiment datasets

The two hyperspectral images are the *Washington DC_HX* dataset and the *Pavia University* dataset. the *Washington DC_HX* image data consists of 191 spectral bands ranging from $400nm$ to $2500nm$ of size 307×1208. The *Pavia University* image data was captured by *Rosis sensor* using 103 spectral bands spanning from $430nm$ to $860nm$, and exhibits a 610×340 pixels with 1.3 meter per pixel geometric resolution. The natural images (RGB) are from the UC Merced Land Use Dataset[2] which are manually extracted from large images from the USGS National Map Urban Area Imagery collection for various urban area around the country including 21 classes and 3 bands. The pixel resolution is 1 foot. The sizes of images are 256×256. The multi-spectral remote sensing images (7 bands)are captured by LandSat-8 satellite, and the full size of the image is 7861×7981 with 7 spectral bands from $430nm$ to $2300nm$.

All the training image patches are randomly cropped from the training part, and then normalized to [0, 1]. Data augmentation with flip and rotation is also done to enlarge the training dataset. The dataset are split into training S_{train} (85%) and validation S_{val} (15%). Table 1 lists the information about the number of patches for training.

For the test data, different images are chosen from outside of the training dataset. The Fig. 4 shows some demo test images in our experiments. The MSI test demo images are cropped from a very large image with the size of 8543×8065. And the part aside from the training dataset of *Pavia University* is chosen as the HSI demo test image.

[1] http://www.pytorch.org

[2] http://weegee.vision.ucmerced.edu/datasets/landuse.html

Table 1 Training samples from each kind of dataset

Dataset	Number of patches
Color image	6300
MSI	4200
HSI	3900
Total	14400

3.2 Parameter tuning

The performance of super resolution reconstruction is closely related to the architecture and the hyper parameters of the network. In our experiments, a large size kernel is utilized in the input layer. For the subsequent convolution layers smaller kernel sizes are chosen.In order to find the optimal network architecture and kernel sizes parameter settings, some different network architectures and parameter settings are discussed in this subsection. Three different network architectures are examined in our experiment. Table 2 shows the results of comparison between different architectures. All the metrics listed in the table are very close,with kernel sizes setting $9 - 7 - 5 - 3$ having the highest value.

Another key factor of determining the speed of converge is the training policy in training phase, which indirectly influence the performance of the reconstruction. The SGD optimizer with initial learning rate 0.01 is employed in experiments. And the learning rate decays every 20 epochs with 200 epochs in the whole training phase totally. In our experiments we find that 200 epochs is sufficient for the training phase. Different initial learning rates are explored in experiments. There are almost no differences when different learning rates are initialized with enough training iterations.

3.3 Comparison performance

To illustrate the performance of the proposed frameworks in super-resolution for remote sensing image data. We compared with state-of-the-art sparse coding based methods SCSR (super-resolution via sparse coding) [10], MHSR (multihypothesis prediction) [5] and bicubic.

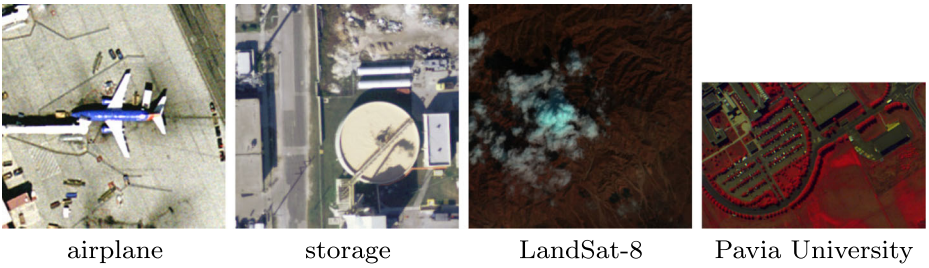

Fig. 4 Visualization of demo test images

Table 2 Performance between different architecture on UC Merced Land Use dataset

Image	Architecture	RMSE	PSNR(dB)	SSIM	time(s)
airplane64	11-5-5-3	0.0204	33.8264	0.9303	2.36
	9-7-5-3	0.0199	34.0164	0.9432	2.75
	7-7-7-3	0.0202	33.8859	0.9325	3.21
freeway45	11-5-5-3	0.0161	35.8544	0.9498	2.31
	9-7-5-3	0.0158	36.0284	0.9622	2.63
	7-7-7-3	0.0160	35.9144	0.9521	3.04
forest50	11-5-5-3	0.0172	35.2653	0.9384	1.98
	9-7-5-3	0.0161	35.8453	0.9531	2.56
	7-7-7-3	0.0169	35.4353	0.9386	2.91

Table 3 Performance comparison of state-of-the-art methods on the UC Merced Land Use Dataset

Images	metrics	bicubic	SCSR	MHSR	SRDCN	DSRDCN	ESRDCN
airplane64	PSNR(dB)	31.7532	32.9205	32.1632	34.0168	34.6562	35.1279
	SSIM	0.9153	0.9467	0.9406	0.9432	0.9474	0.9531
	RMSE	0.0258	0.0195	0.0222	0.0199	0.0185	0.0175
	time(s)	-	485.96	284.93	2.75	3.59	5.76
freeway45	PSNR(dB)	33.3892	33.3694	32.9025	36.0287	36.5041	37.1158
	SSIM	0.9430	0.9611	0.9613	0.9622	0.9633	0.9650
	RMSE	0.021406	0.0160	0.01745	0.0158	0.0149	0.0139
	time(s)	-	292.37	238.26	2.63	3.45	5.36
forest50	PSNR(dB)	33.8991	34.1293	33.8204	35.8456	35.7542	36.0138
	SSIM	0.9191	0.9483	0.9454	0.9531	0.9494	0.9546
	RMSE	0.0201	0.0165	0.0171	0.0161	0.0163	0.0158
	time(s)	-	439.17	279.79	2.56	3.31	5.26

Table 4 Performance Comparison on the Lendset8 multi-spectral image data

Images	metrics	bicubic	SCSR	MHSR	SRDCN	DSRDCN	ESRDCN
Region1	PSNR(dB)	39.5833	40.2012	40.8785	41.0031	41.1083	41.2643
	SSIM	0.9867	0.9844	0.9902	0.9905	0.9913	0.9921
	RMSE	0.01049	0.0097	0.0090	0.0089	0.0088	0.0086
	time(s)	-	769.30	2106.08	31.96	42.46	65.76
Region2	PSNR(dB)	37.2321	39.0283	38.6904	38.9945	39.0631	39.1907
	SSIM	0.9772	0.985	0.9841	0.9852	0.9858	0.9901
	RMSE	0.0138	0.0112	0.0116	0.0112	0.0111	0.0110
	time(s)	-	266.15	529.38	23.81	30.56	49.31

3.3.1 Experiments on nature and multi-spectral images

Table 3 shows results of comparison on the UC Merced Land Use Dataset. Table 4 lists the performances on the LandSat-8 remote sensing images. Bicubic is treated as a standard reconstruction algorithm, elapsed time is not computed and considered as a metric in

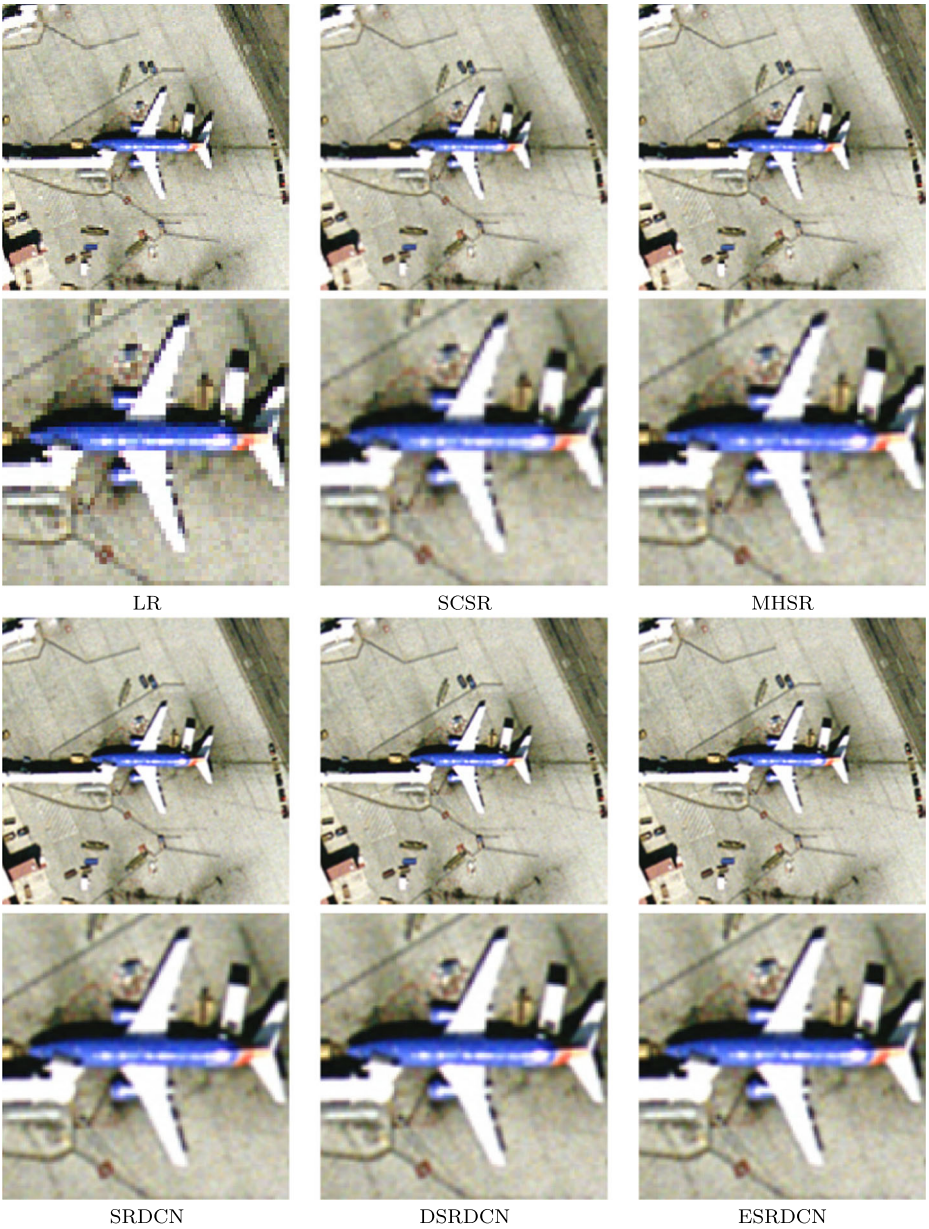

Fig. 5 Visualization performance on natural image of airplane

our experiments. Our proposed method outperform all the other methods. The CNN based methods shows better results with all metrics over SCSR and MHSR, which are dictionary learning based methods. The CNN based methods have higher PSNR, SSIM value and lower RMSE value, and, noticeably, much less computational time. And, among all the methods ESRDCN gets the best performance. The Figs. 5 and 6 show the input LR images

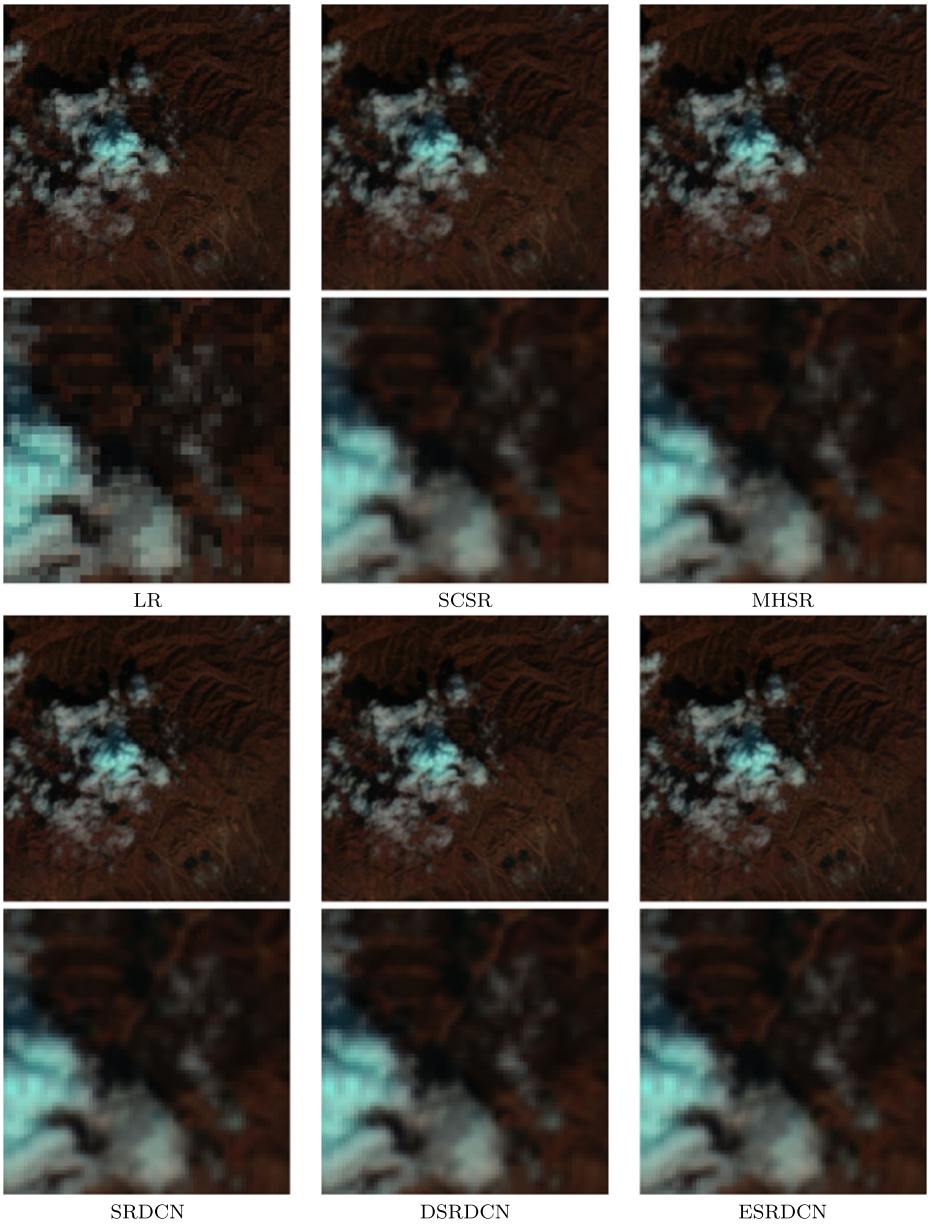

Fig. 6 Visualization performance on multi-spectral image by pseudo color, with NIR, red, green bands for red, green, and blue colors respectively

Table 5 Performance Comparison on hyperspectral image data

Images	metrics	bicubic	SCSR	MHSR	SRDCN	DSRDCN	ESRDCN
Washington DC (191 bands)	PSNR(dB)	34.4889	34.9437	35.4424	35.6303	35.7441	35.9016
	SSIM	0.9342	0.9473	0.9501	0.9507	0.9514	0.9546
	RMSE	0.0189	0.0179	0.0169	0.0165	0.0163	0.0160
	time(s)	-	12098.55	21731.06	462.76	513.49	761.21
Pavia University (103 bands)	PSNR(dB)	32.4978	34.3425	33.8107	34.8775	35.0902	35.2103
	SSIM	0.9122	0.9423	0.9374	0.9564	0.9592	0.9623
	RMSE	0.0237	0.0191	0.02039	0.0180	0.0176	0.0174
	time(s)	-	10763.81	24207.944	264.43	375.46	538.94

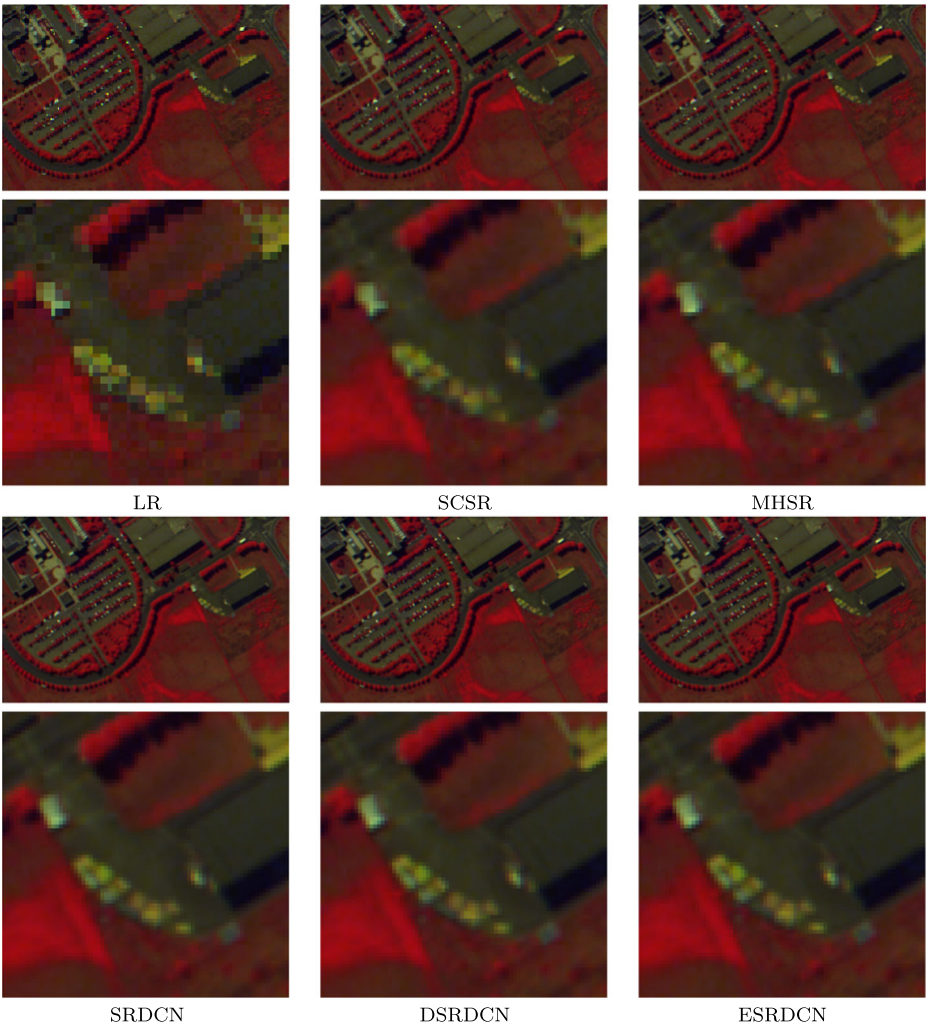

Fig. 7 Visualization performance on hyperspectral image by pseudo color, with NIR, red, green bands for red, green, and blue colors respectively

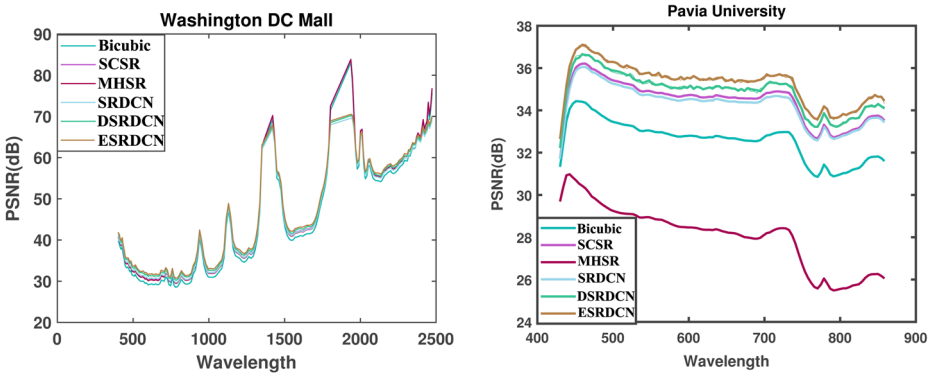

Fig. 8 The PSNR curves as the functions of the wavelength of spectral bands for the test methods onhyperspectral image dataset

and outputs of all the experiment methods. From Fig. 5, better results can be seen with CNN based methods, the edges are more clear, especially the edges around the wings of the airplanes. From what shows in Fig. 6, the proposed CNN based methods have more detailed textures,both at the cloud areas and the mountainous areas.

3.3.2 Experiments on hyperspectral images

Same results came out from the experiments on hyperspectral image data. Table 5 shows numeric results of the whole test data, and Fig. 7 shows the image results. The CNN based methods show their prevalence in edge preservation and shape restoration, with ESRDCN having the best numeric results. As for the computational cost, the proposed CNN based methods are more than 10 times faster than the SCSR and MHSR. That is because CNN based methods only consist multiply and addition operations in the inference procedure, while dictionary based methods need to find the optimal sparse parameter.

Figure 8 illustrates performance of PSNR of each band, the PSNR of ESRDCN is persistently higher than the other methods at all bands. Two visible peaks exist in the spectral wavelength of $1400 - 1550nm$ and $1700 - 2000nm$ with the PSNR gain curves reaching $80dB$ on hyperspectral dataset $WashintonDC$, which is an abnormally large value. After further investigation, we discover that the tested data values in these spectral bands are close to zero. These unusual values result in unreasonable PSNR curves.

Classsification accuracies are also provided on $WashintonDC$ and $Pavia$ datasets. The classification accuracy is used to further test the applicabilities of the image results. Tables 6 and 7 gives the classification results with basic SVM classification method with 100 training

Table 6 Results of classifications on Pavia University with different super resolution methods via SVM

Metrics	ESRDCN	DSRDCN	SRDCN	MHSR	SCSR	Bicubic
OA	94.88%	94.55%	94.57%	94.19%	93.33%	92.07%
AA	94.17%	93.30%	93.59%	93.23%	93.60%	92.33%
Kappa	0.9426	0.9390	0.9250	0.9250%	0.9250	0.9106

Table 7 Results of classifications on Washington DC with different super resolution methods via SVM

Metrics	ESRDCN	DSRDCN	SRDCN	MHSR	SCSR	Bicubic
OA	80.44%	79.96%	78.61%	77.21%	76.41%	73.90%
AA	60.12%	59.82%	56.72%	57.61%	58.46%	61.97%
Kappa	0.7632	0.7583	0.7632	0.7227%	0.7110	0.6737

samples. From the results, the CNN based methods all have higher overall accuracy (OA), average accuracy (AA) and kappa coeffficient, with ESRDCN having the highest values.

4 Conclusions

In this paper, three different types of deep convolution networks for super-resolution were proposed and applied on remote sensing image data. The proposed SRDCN learns an end-to-end mapping function to reconstruct a high-resolution image from its low-resolution version with pre- and post-processing. Motivated by residual learning, an improved network addresses learning the tiny differences between the HR and LR images which make it possible for the network to go deeper and recover image with better quality. Besides, a multi-scale version was also utilized to image super resolution. The proposed methods were validated on three different types of remote sensing datasets and compared with some state-of-the-art methods, experimental results demonstrated that the proposed methods could provide better performance. Since the architecture of proposed methods are simply consisted of convolution operation, the proposed methods are much faster than dictionary based methods. And, the consistent prevalence of the proposed methods on different kinds of data also proves their robustness.

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